

Edge AI Patent Assessment

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SUMMARY

Conclusion: Label X: TRL 85.0 / Claim 85.0
Key reasons: TRL readiness score 85.0 / Claim breadth score 85.0
Purpose: Edge AI █████-████ ███ | Keywords | CPC | Period N/A ~ N/A
Top-line: N=2 / M=None / K=0 | Label X

Recommendations

- FTO █████ (owner IP █, due +7d): █████ █████
- █████ █████ (owner █████ █, due +14d): SEP █████ █████
- █████ █████ (owner AI █████, due +21d): █████ █████ 10█ █████

Risks

- FTO_risk: █████ ███ / █████ █████
- █████ █████: 0/10 █████ █████

Trust meta: missing 100.0% | Executed 2025-10-23T10:33:25.145216+09:00 | Policy v0

MAIN BODY

1. Scope & Method

- scope: {'keywords': '', 'cpc': '', 'period': 'N/A ~ N/A', 'target_market': 'trl'}
- method: {'normalisation': '█████ → 5█(████ ███)', 'weights': '███/███/███ █████', 'label_rule': 'A/P/M/X █████ FTO █████'}

2. Dataset Overview

- documents: 2
- quality: {'missing_metrics': ['AGE_YEARS', 'CITATIONS_TOTAL', 'COUNTRIES', 'FAMILY_SIZE', 'GENERALITY', 'HHI', 'LITIGATION_FLAG', 'ORIGINALITY', 'RENEWAL_STATUS', 'SEP_INDICATION'], 'translation_flags': 'N/A', 'dedup': 'N/A'}

3. Core Metrics

- trl_score: 85.0
- claim_score: 85.0

4. Edge Fitness

5. Competitive & Standards

6. Risk Profile

- flags: FTO_risk
- data_gaps: AGE_YEARS, CITATIONS_TOTAL, COUNTRIES, FAMILY_SIZE, GENERALITY, HHI, LITIGATION_FLAG, ORIGINALITY, RENEWAL_STATUS, SEP_INDICATION

7. Ranking & Labelling

- total_score: 66.44736842105263
- label: X
- distribution: {'A': 0, 'P': 0, 'M': 0, 'X': 1}

8. Decision Cards

- title=WO2018097365A1; owner=...; decision=Monitor; signals=['TRL readiness score 85.0', 'Claim breadth score 85.0']; risks=[{'item': 'FTO_risk', 'impact': '████ ███ / █████ █████'}, {'item': '████ █████', 'impact': '0/10 █████ █████'}]

9. Conclusion

- TRL readiness score 85.0
- Claim breadth score 85.0

REFERENCE

- [R1] NBER Working Paper 7741 - <https://www.nber.org/papers/w7741>
[R2] WIPO IP Indicators - <https://www.wipo.int/web-publications/world-intellectual-property-indicators-2024-highlights/en/patents-highlights.html>
[R3] USPTO Official Gazette - <https://www.uspto.gov/web/offices/com/sol/og/2012/week52/TOCCN/item-80.htm>
[R4] ETSI SEP List - https://www.etsi.org/deliver/etsi_sr/000300_000399/000314/02.32.01_60/sr_000314v023201p.pdf
[R5] DOJ / FTC Guidelines - <https://www.justice.gov/atr/herfindahl-hirschman-index>
[R6] NBER Working Paper 8498 - <https://www.nber.org/papers/w8498>
[R7] Google Patents - <https://patents.google.com/patent/WO2018097365A1>
[R8] Local PDF - data\pdfs\WO2018097365A1.pdf
[R9] Local PDF - data\pdfs\SWTRLs.pdf

APPENDIX

purpose: Edge AI ■■■ ■■■■.■■■■ ■■■
pipeline: PDF → Chunk → Evidence RAG → ■■■ ■■■■
formulae: ■■■■ = $\Sigma(\text{metric} \times \text{weight})$, ■■■ ■■■ ■■■■
label_policy: Adopt/Prototype/Monitor/Archive (■■■ ■)
device_thresholds:
sensitivity: contributions={'trl': 0.2631578947368421, 'claim': 0.2631578947368421, 'legal': 0.10526315789473684, 'family': 0.13157894736842105, 'foreign': 0.10526315789473684, 'renewal': 0.07894736842105263, 'std': 0.05263157894736842}, redistributed=True, missing_keys=['generality', 'originality'], fto_penalty=0.0
risk_policy: ■■■■ ■■■/■■■ ■■■■ ■■■■ ■■■■ ■■■
execution_meta: timestamp=2025-10-23T10:33:25.145216+09:00, version=v0

EVIDENCE

TRL [Local RAG]

TRL level: 8.5

Example progression: **prototype** -> **pilot** -> field test -> **deployed production** services.

An **on-device** inference apparatus is described, transitioning a **prototype** into a **pilot** deployment with **real-time** quantized processing for edge AI scenarios.

TRL 5 corresponds to validating a software **prototype** in a relevant environment. TRL 6 and TRL 7 involve field test pilots with partially **deployed** systems. TRL 8 and TRL 9 indicate production-ready and fully **deployed** software in an operational environment.

CLAIMS [Local RAG]

Independent claims: 1, Avg tokens: 177.0

A device **comprising** an **on-device** processor configured to execute a quantized neural network in **real-time** within an operational environment, the device including telemetry for **prototype** to **pilot** upgrades, adaptive calibration pipelines, secure enclave management, federated aggregation buffers, and a plurality of edge accelerators coordinated by a policy engine that maintains latency thresholds below ten milliseconds while preserving power budgets across mission profiles. The device further comprises memory partitions, compression controllers, wireless radios, fallbacks for disconnected operation, proactive failure detection routines, and orchestrated deployment hooks that enable continuous integration of model updates validated through field trials, **pilot** deployments, and **production** readine

The device of claim 1 **wherein** the processor is configured to execute federated updates on-device, dynamically batching encrypted gradients, reconciling conflict resolution policies, and orchestrating synchronization checkpoints. The device of claim 2 **wherein** the memory subsystem provides at least one secure enclave with attestation pipelines, redundancy failover, and audit telemetry forwarding pathways.

API_METRICS [Local RAG]

SCORE [Local RAG]